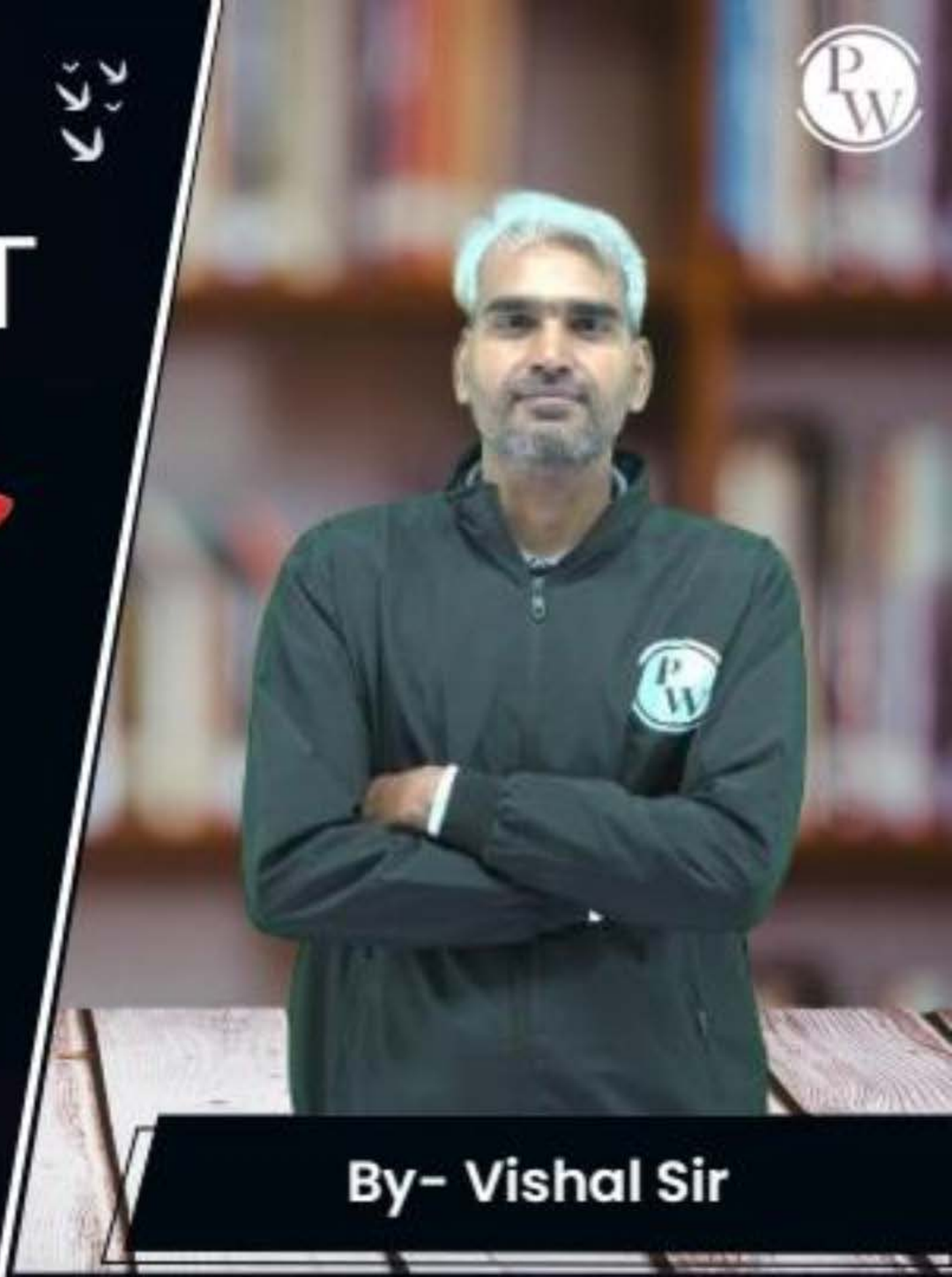
Computer Science & IT



Database Management System

Entity Relationship Model & Integrity constraints

Lecture No. 01



By- Vishal Sir

Recap of Previous Lecture



Topic B+ Tree Index



Slide

Topics to be Covered



Topic — ER model & ER diagram

Topic — Relational model & Integrity constraints

Slide



Topic : Entity-Relationship Model (ER-Model)

- ER Model is a high-level data model diagram which defines the conceptual view of the database (OR) that gives the graphical representation of the logical structure of the database.

- Entity-Relationship model is based on objects called entities, and relationship among these entities.



Topic : Entity

- An entity refers to any object having either a physical existence or a conceptual existence. For example, a school, a person, a house, a university, a company or a job.



Topic : Entity Set

To identify an entity uniquely we use attributes.

An entity set is a collection or set of all entities of a particular entity type at any point in time.

Entity type (or Entity set)

Student

Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19

This complete table is referred to as "Student Entity Set" and each row represents an "entity".

These are entities



Topic : Entity Set

An entity set is represented by a rectangle in ER diagram



← Student entity set.

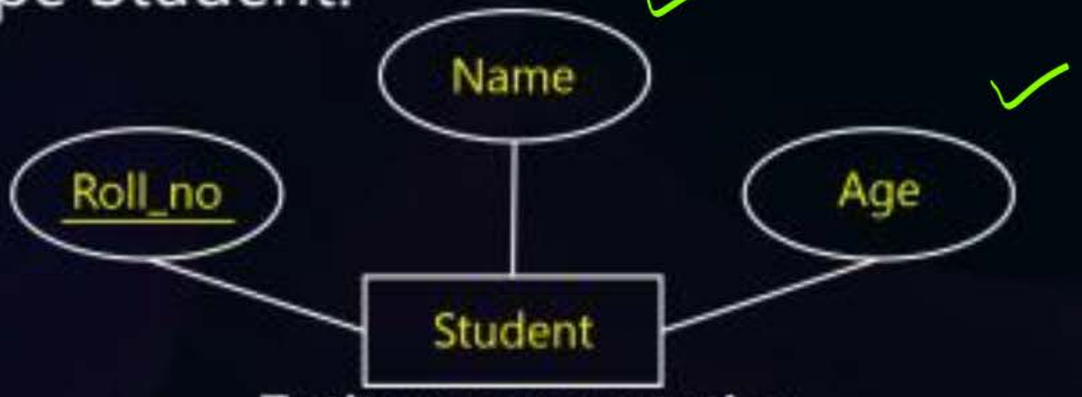


Topic : Attribute

Each entity is identified uniquely by the values of its attributes.

An attribute in an Entity-Relationship Model describes the properties or characteristics of an entity. It is represented by an oval or ellipse shape in the ER diagram. Every oval shape represents one attribute and is directly connected to its entity set which is rectangle in shape.

For example, roll_no, name, and age are the attributes which define entity type Student.



Entity representation

Roll_no	Name	Age
1	Ram	20
2	Shyam	19
3	Mohan	20
4	Arjun	19



Topic : Types of Attributes

i. Simple attribute

ii. Composite attribute

iii. Single-valued attribute

iv. Multi-valued attribute

v. Key attribute

vi. Derived attribute

vii. Partial key attribute



Topic : Types of Attributes

(i) Simple Attributes: An attribute which contains an atomic value and cannot be divided further is called a simple attribute. Example, Roll_no, Sex, etc.

(ii) Composite attribute: An attribute which can further be divided. It is an attribute which is composed of several components. Example name, address, etc.

first_name Middle_name last_name

(iii) Single-valued attribute: An attribute which can have only one value.

(iv) Multi-valued attribute : An attribute which has multiple values (more than one value)

(v) Key attribute : An attribute which can identify all entities uniquely.

(vi) Derived attribute : An attribute whose value can be derived or found from any other attribute is known as derived attributes.

(vii) Partial key attribute : An attribute which can identify a group of entities.



Topic : Types of Attributes



Simple Attribute



Multivalued Attribute



Composite Attribute



Key Attribute



Partial Attribute



Derived Attribute

associated with Weak entity set.



Topic : Relationship in ER model

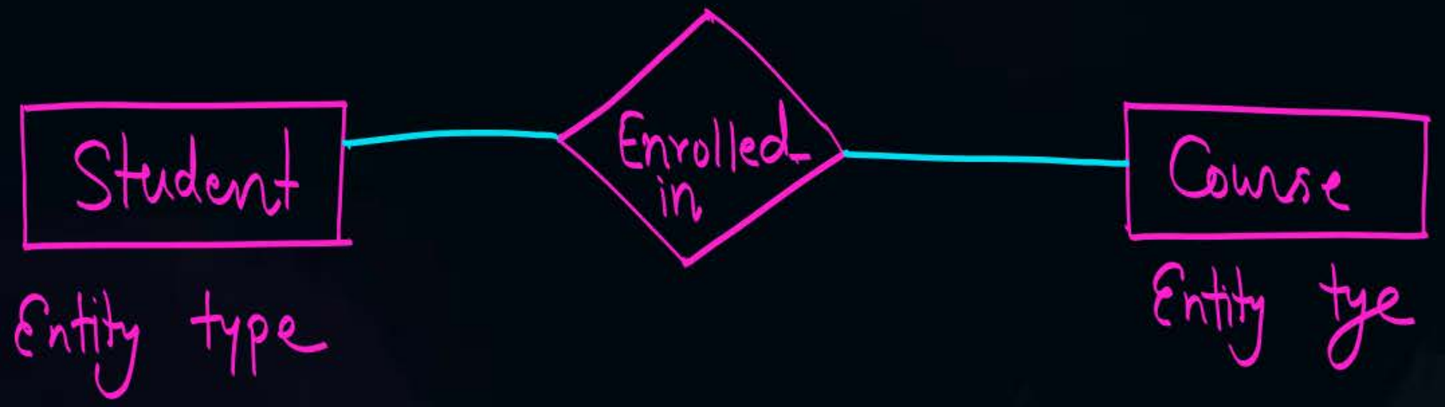
A Relationship Type represents the association between entity types. For example, 'Enrolled' is a relationship type that exists between entity type Student and Course. In ER diagram, the relationship type is represented by a diamond and connecting the entities with lines.

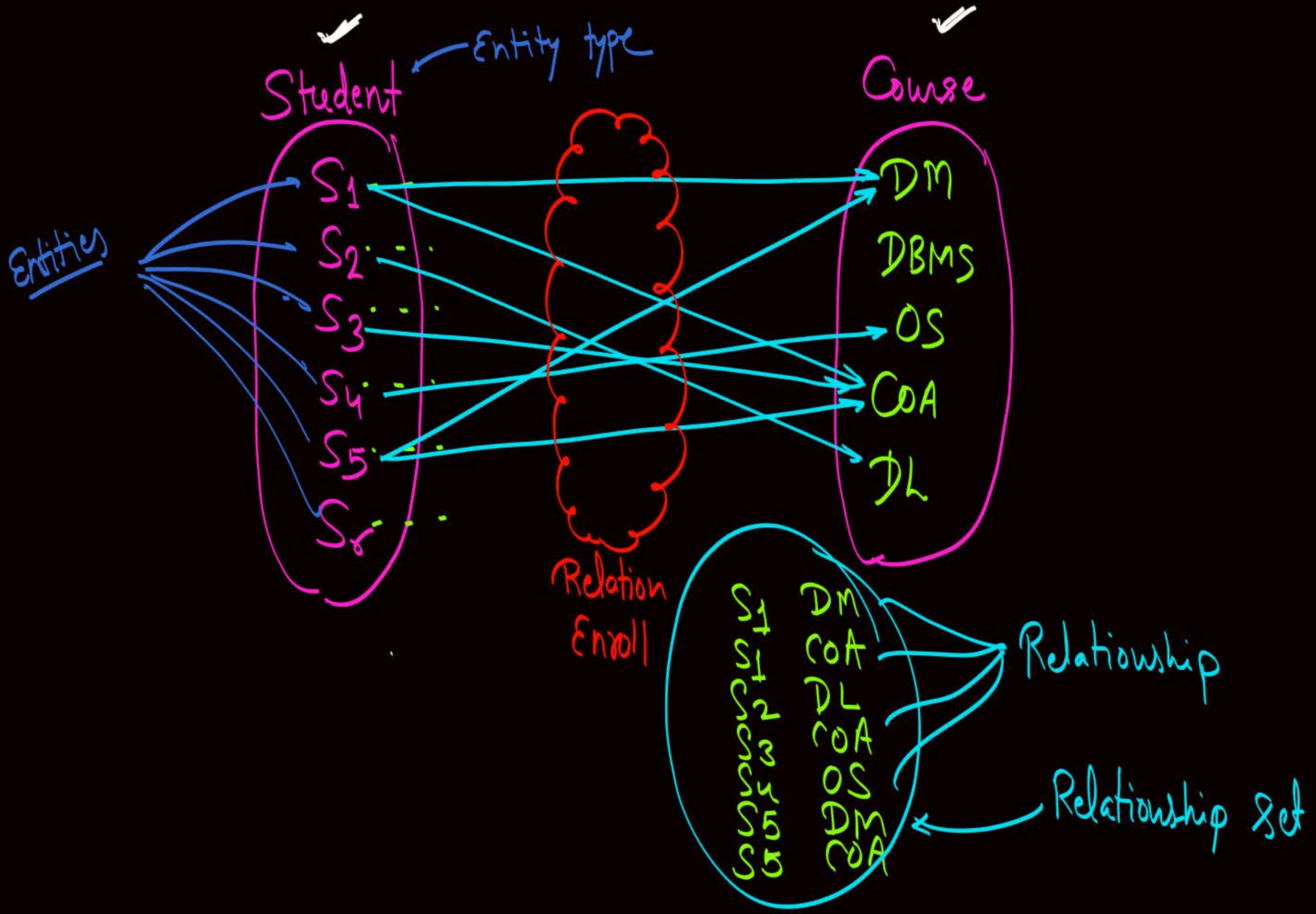




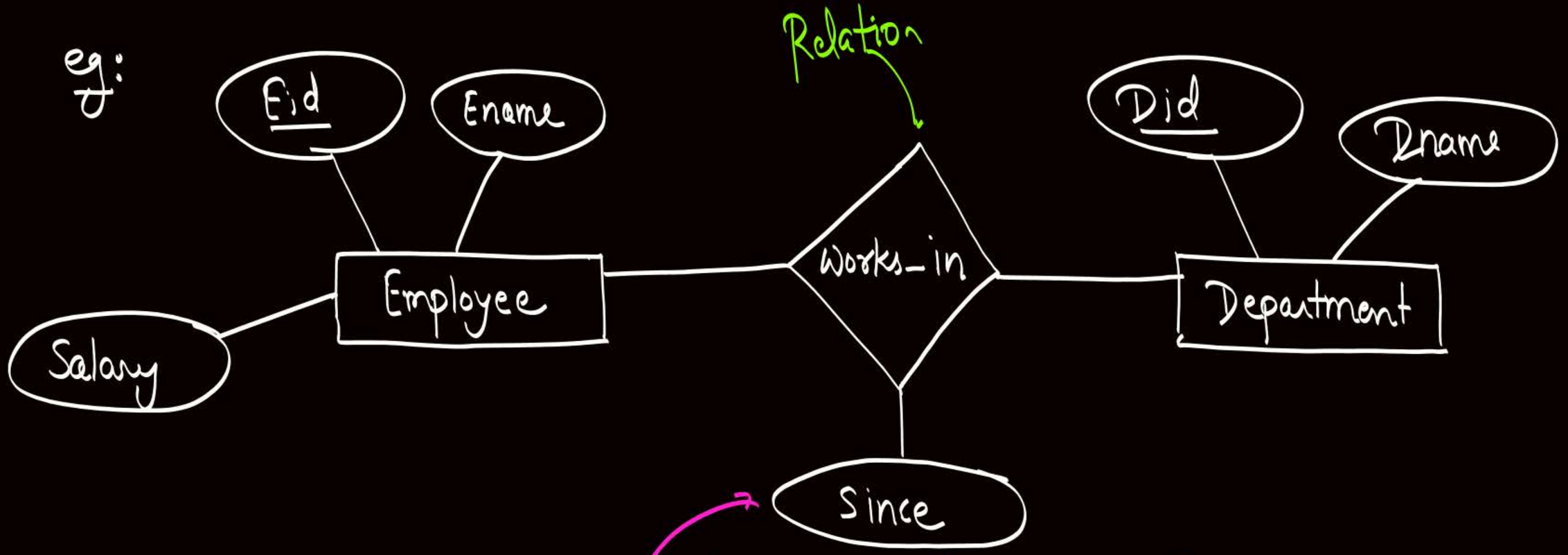
Topic : Relationship in ER model

Example:




Entity type
Student
Course
Entities
S1
S2
S3
S4
S5
Sn
DM
DBMS
OS
CoA
DL
Relation
Enroll
S1 DM
S2 CoA
S3 DL
S4 CoA
S5 OS
S5 DM
S5 CoA
Relationship
Relationship Set

eg:



it is called descriptive attribute

attribute of relation

↳ it will specify that since when an Employee works_in a Particular department

{ It is not necessary for all relations to have their own attributes



Topic : Descriptive attribute

Descriptive attributes are use to provide the information related to relationship set.



Topic : Degree of a Relationship

The number of different entity sets participating in a relationship is called the degree of a relationship.

Degree of a relationship = Number of entity sets participating in a relationship



Topic : Types of Relationship

On the basis of degree of a relationship set, a relationship set can be classified into the following types-

i. Unary Relationship: Unary relationship set is a relationship set where only one entity set participates in a relationship set.

ii. Binary Relationship: Binary relationship set is a relationship set where two entity sets participate in a relationship set.

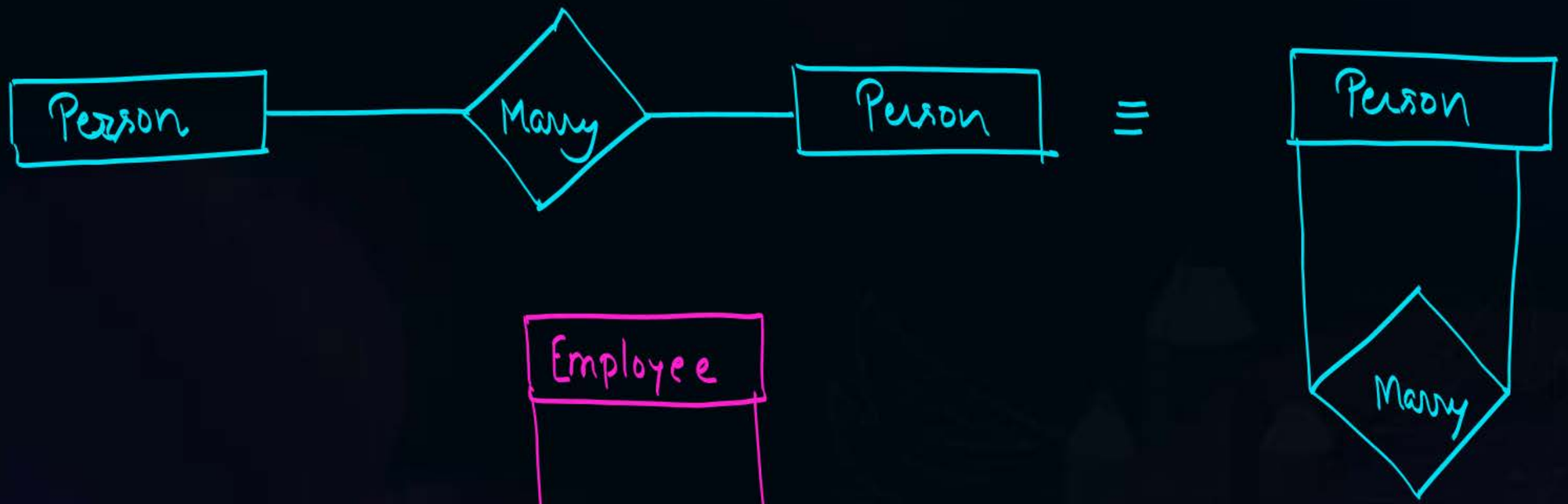
iii. Ternary Relationship: Ternary relationship set is a relationship set where three entity sets participate in a relationship set.

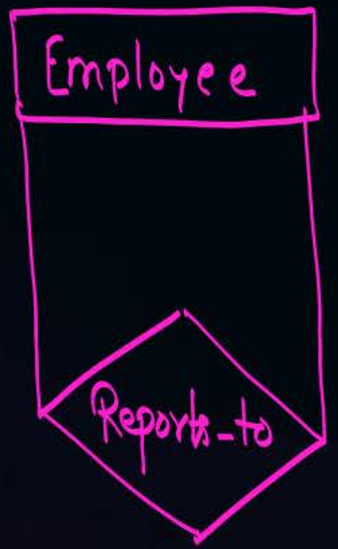
iv. N-ary Relationship: N-ary relationship set is a relationship set where 'N' entity sets participate in a relationship set.



Topic : Unary Relationship

Eg:







Topic : Binary Relationship

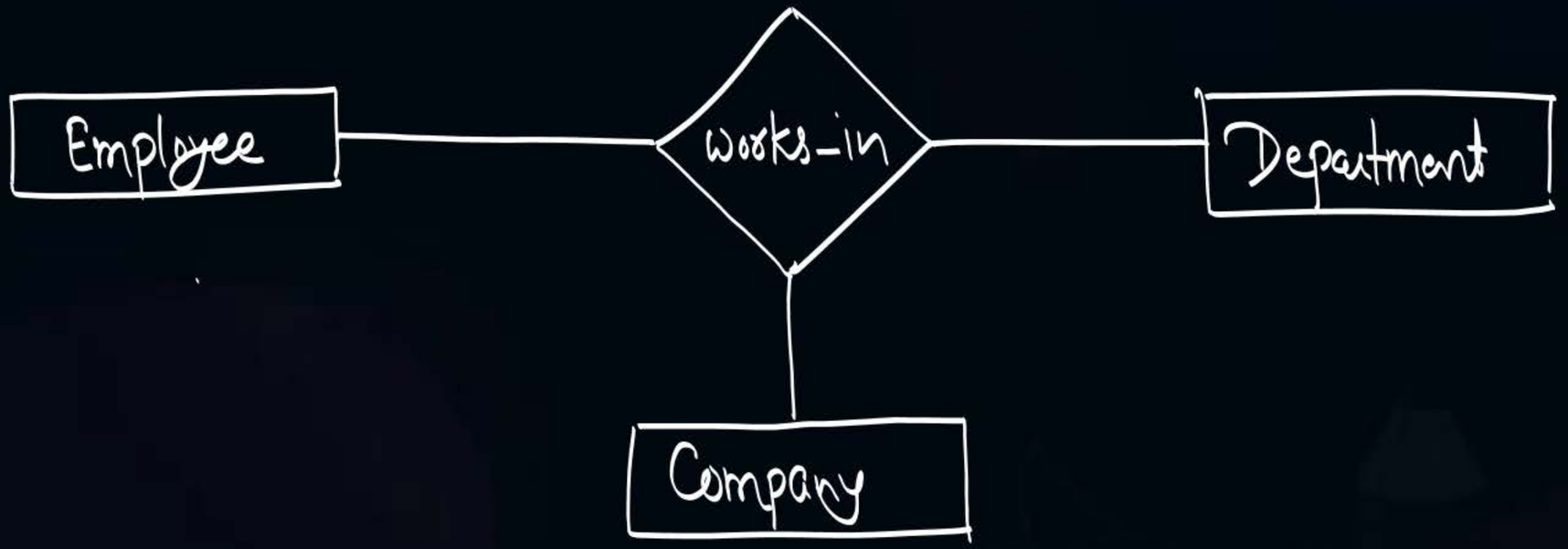
eg.





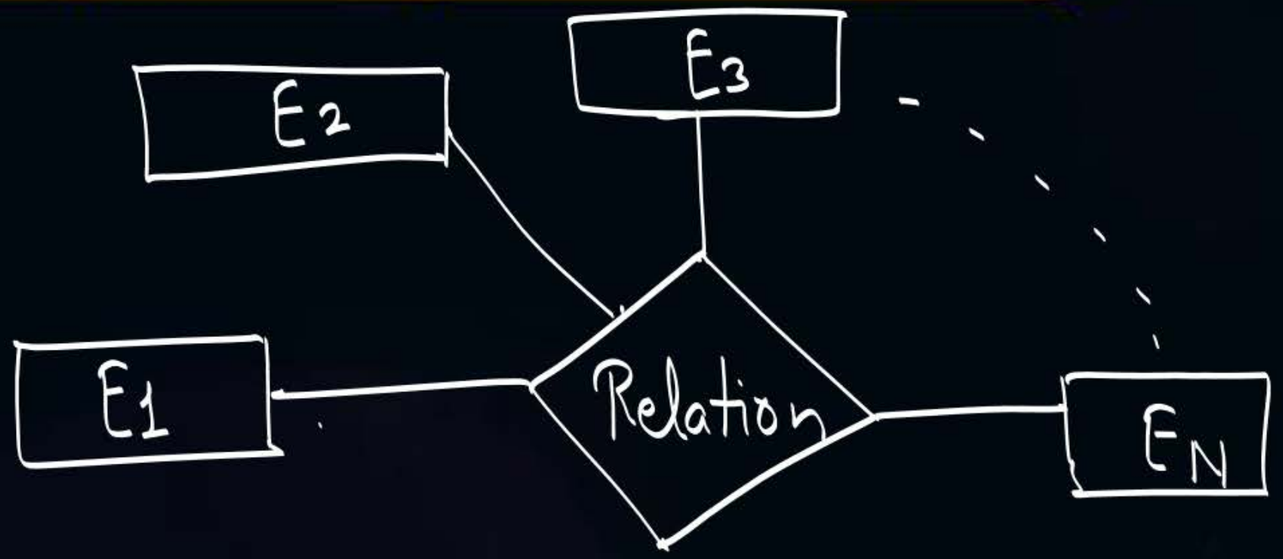


Topic : Ternary Relationship





Topic : N-ary Relationship





Topic : Participation constraints

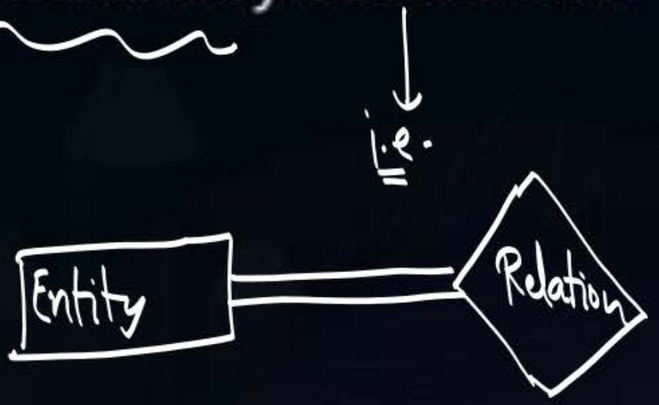
There are two types of participation constraints :

1. Total participation
2. Partial participation

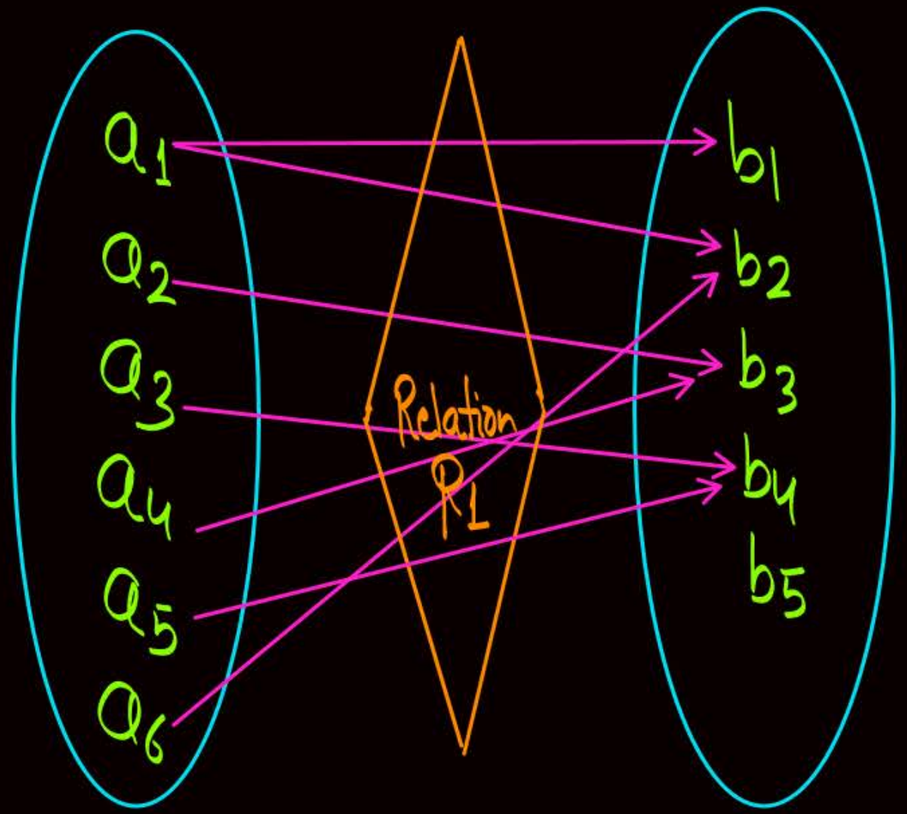


Topic : Total participation

- If every entity of an entity set participate in the relationship set, then that entity set is said to have total participation.
- Total participation of an entity set in a relationship set is denoted by double line



Entity set → E_1 E_2 ← entity set



Every entity of entity set E_1 is participating in Relationship set R_1

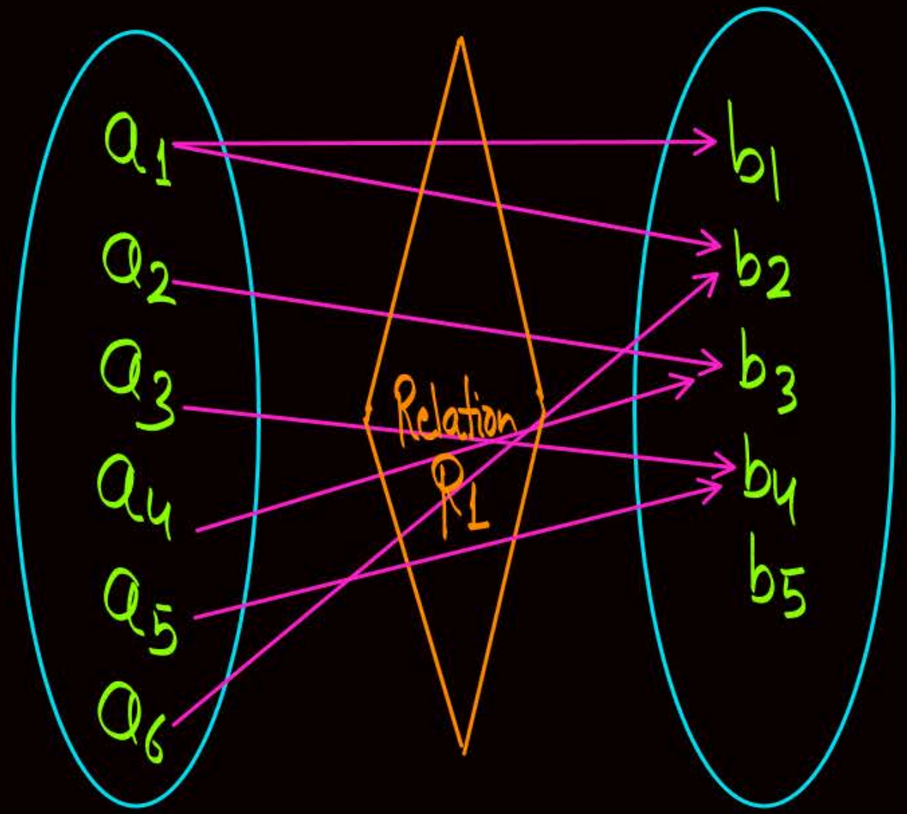
∴ E_1 has total participation w.r.t. relationship set R_1

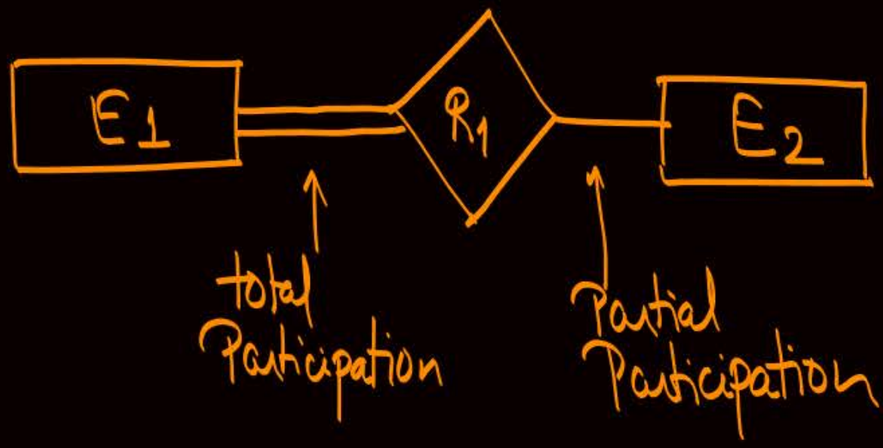


Topic : Partial participation

- If at least one entity of entity set is not participating in the relationship set, then that entity set is said to have partial participation.
- Partial participation of an entity set in a relationship set is denoted by single line.

Entity set → E_1 E_2 ← entity set



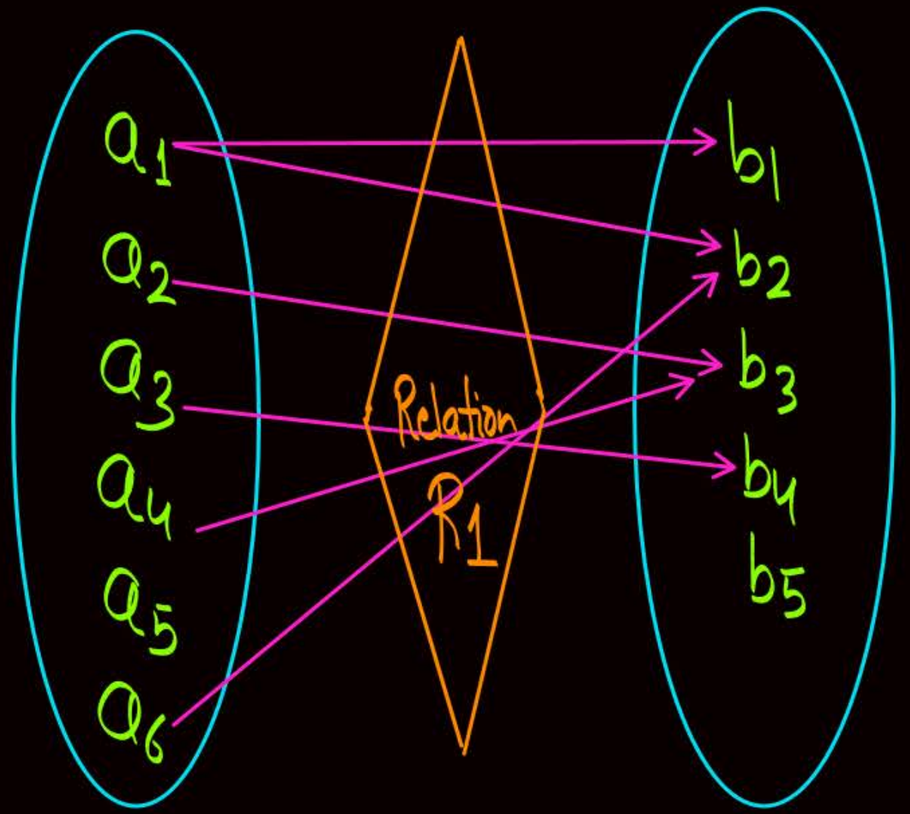


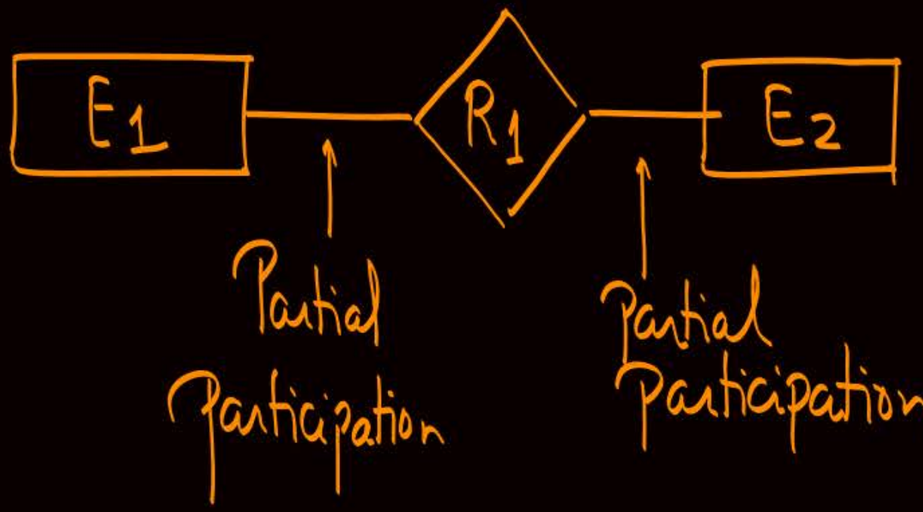
Entity b_5 of entity set E_2 is not participating in Relationship set R_1

∴ E_2 has partial participation w.r.t. relationship set R_1

Entity set → E_1

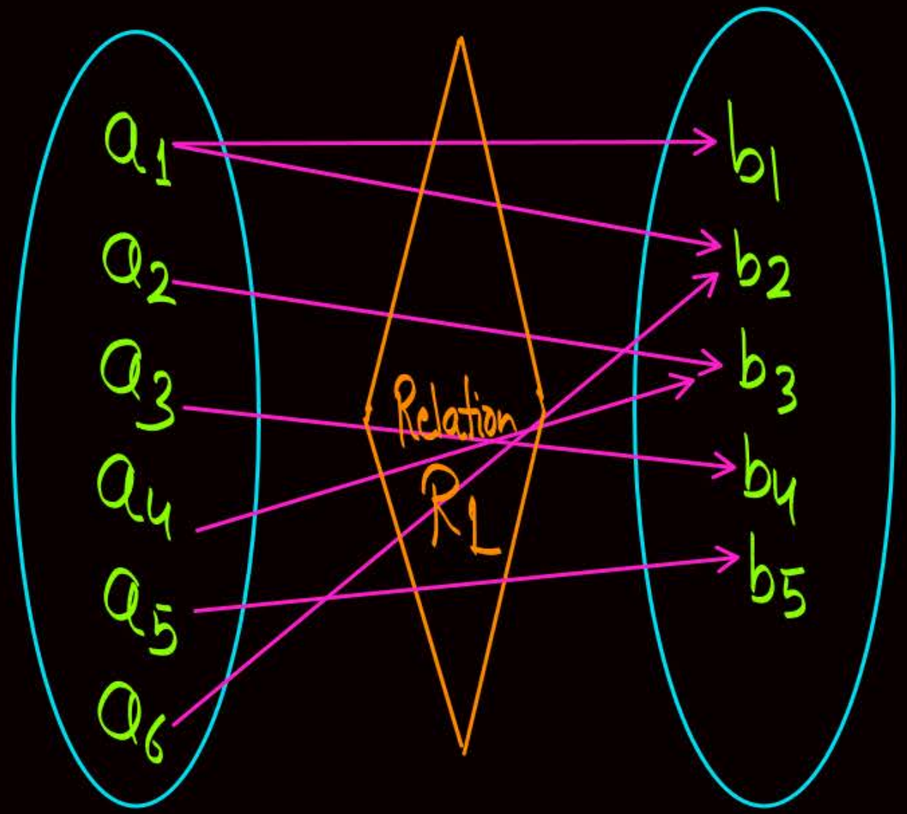
E_2 ← entity set

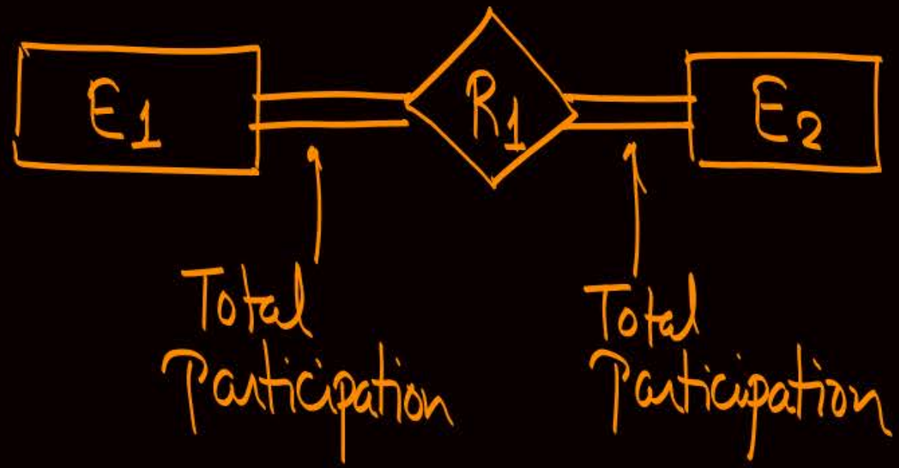




Entity set → E_1

E_2 ← entity set







Topic : Mapping cardinality

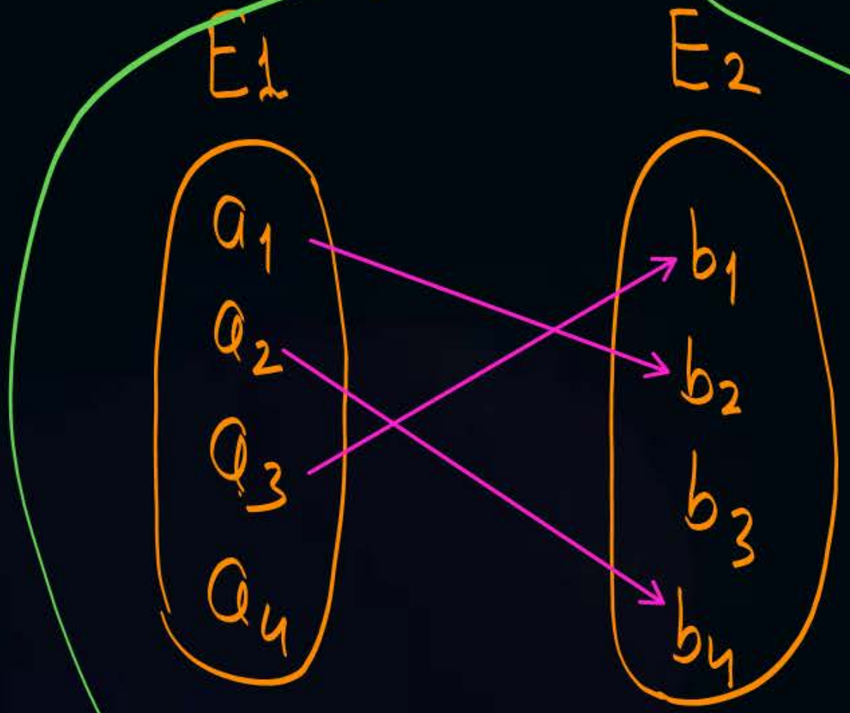
Mapping cardinality or cardinality ratio is used to denote the number of entities to which another entity can be associated through a certain relation set.

For a binary relationship test mapping cardinalities must be one of the following types:

1. One-to-one (1 : 1)
2. One-to-many (1 : N)
3. Many-to-one (N : 1)
4. Many-to many (M : N)



Topic : One-to-one

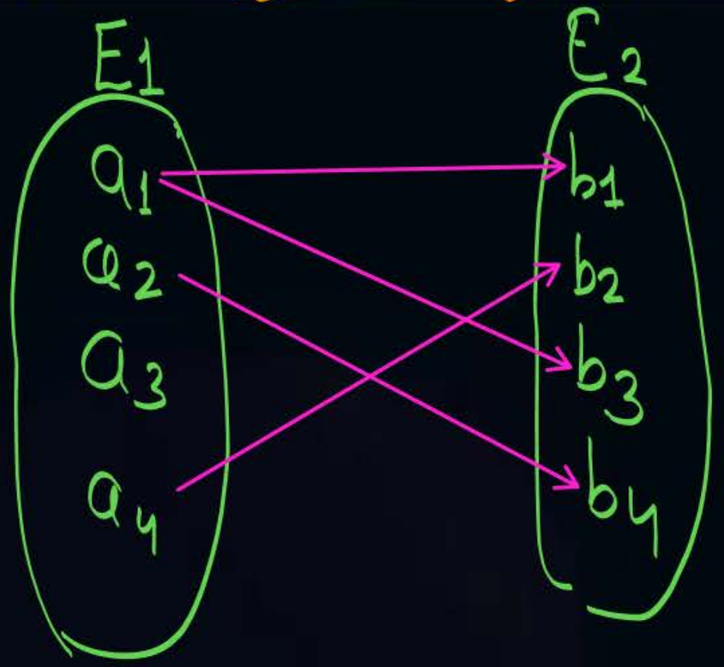


One entity of first entity set Can associate with at most one entity of entity set E_2

And One entity of entity set E_2 Can associate with at most one entity of entity set E_1



Topic : One-to-many



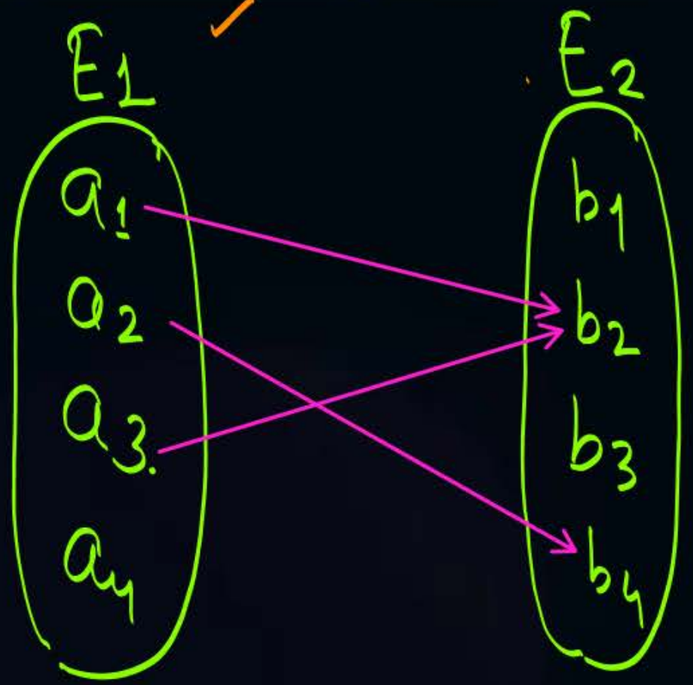
One entity of entity set E_1 can associate with many (0 or more) entities of entity set E_2

and

One entity of entity set E_2 can associate with at most one (0 or 1) entity of entity set E_1



Topic : Many-to-one



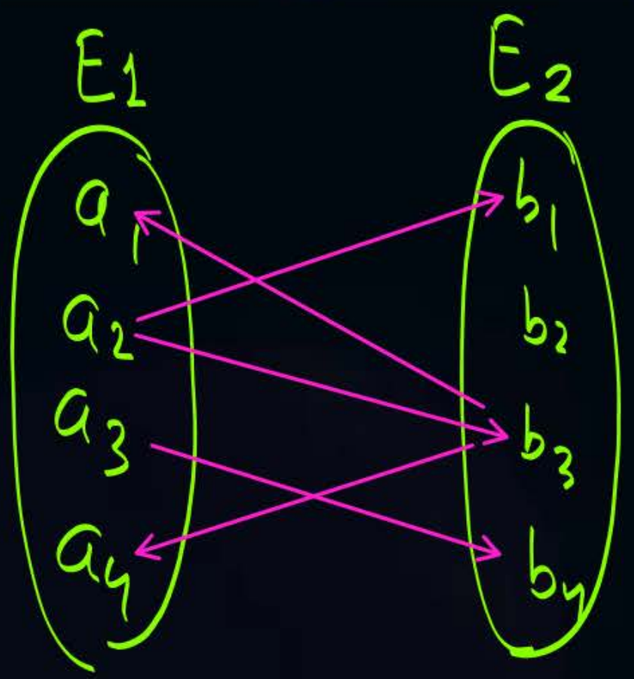
entities of set E_1 can associate with at most one entity of set E_2

and

One entity of set E_2 can associate with many (0 or more) entities of set E_1



Topic : Many-to-many

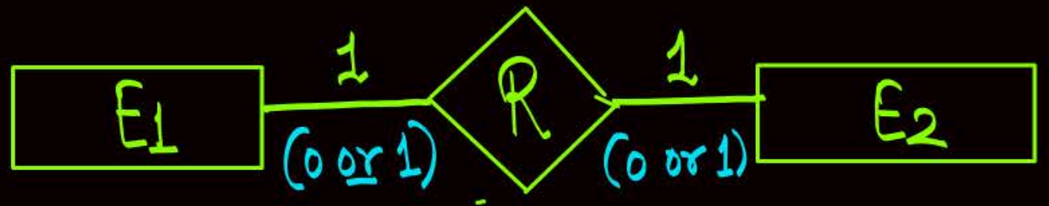


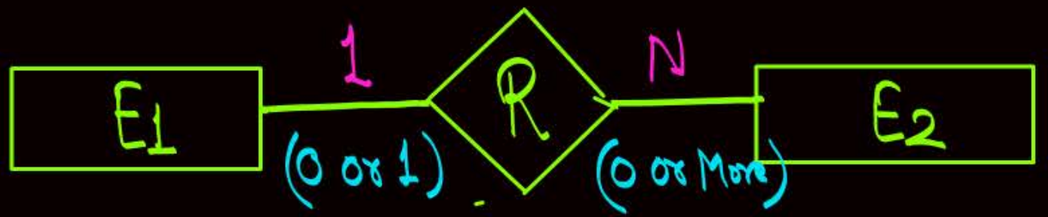
Any number of entities of set E_1 can associate with any number of entities of set E_2

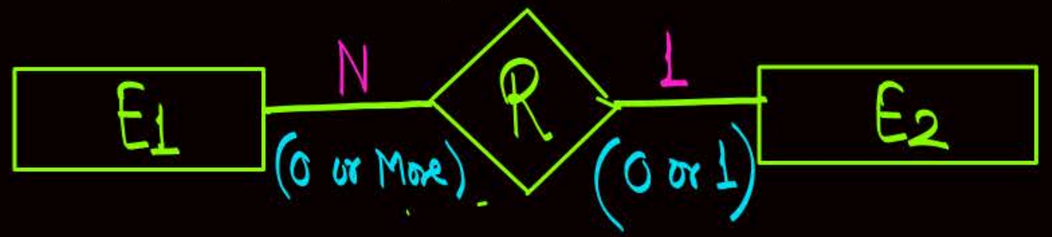
entity of Entity set E_1 can associate with many (0 or more) entities of Entity set E_2

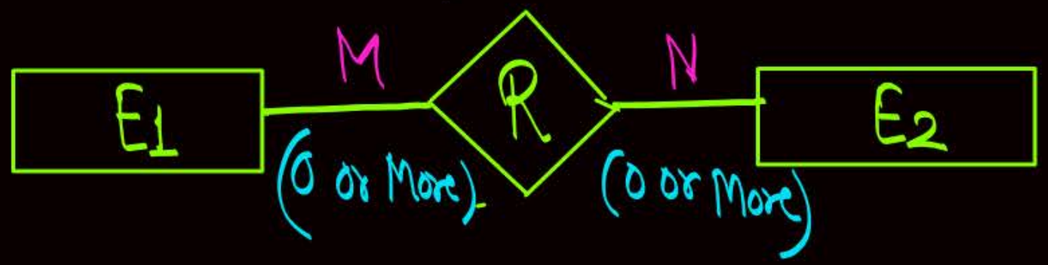
&

Entity of entity set E_2 can associate with many (0 or more) entities of Entity set E_1

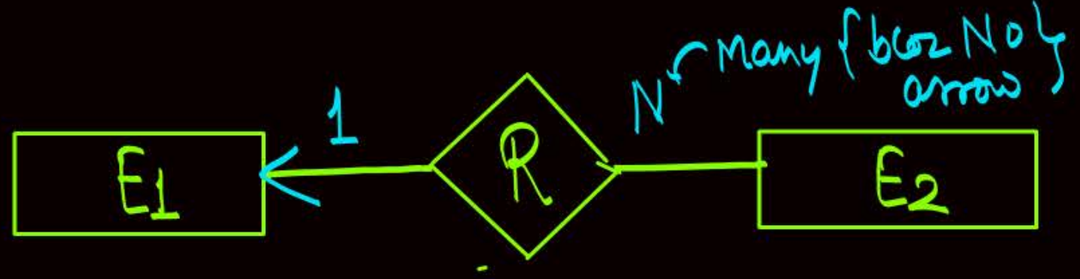

E1
1
(0 or 1)
R
1
(0 or 1)
E2

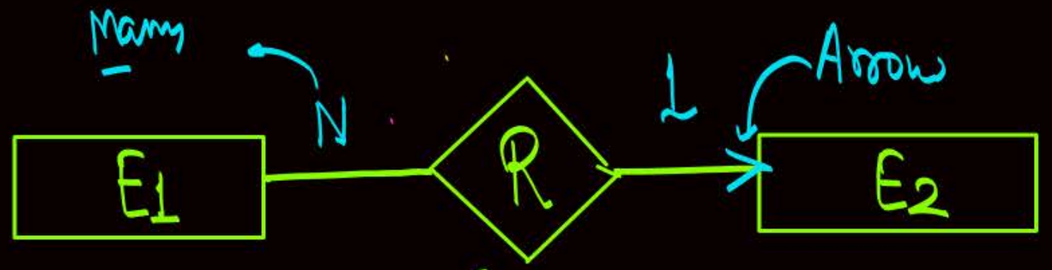

E1
1
(0 or 1)
R
N
(0 or More)
E2

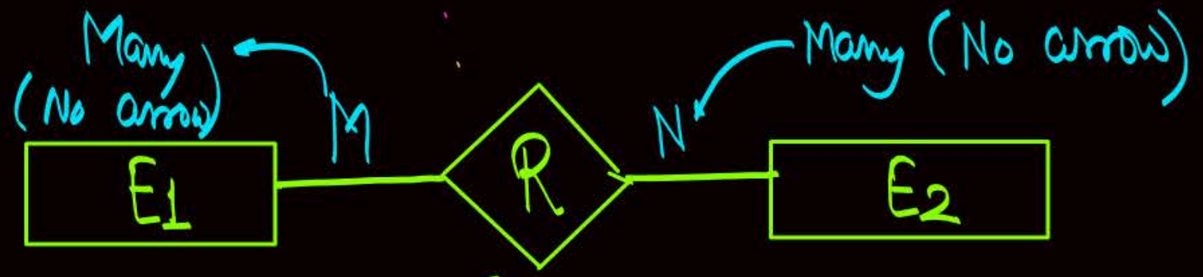

E1
N
(0 or More)
R
1
(0 or 1)
E2

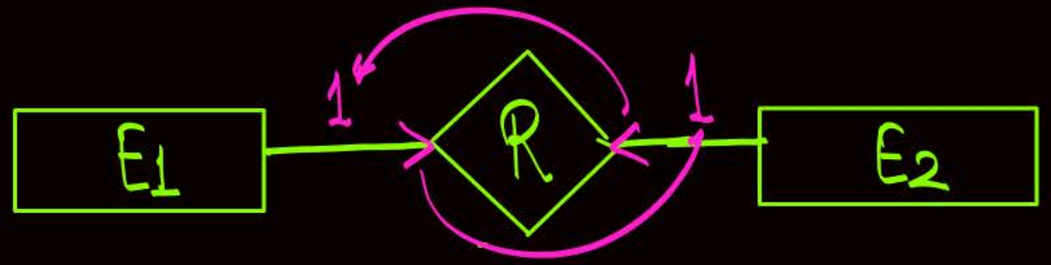

E1
M
(0 or More)
R
N
(0 or More)
E2


E1
R
E2
1
1


E1
R
E2
1
N
Many (bcz No arrow)


Many
N
E1
R
1
Arrow
E2


Many
(No arrow)
M
E1
R
N
Many (No arrow)
E2


E1
1
R
1
E2


E1
1
R
N
E2


E1
N
R
1
E2


E1
M
R
N
E2



Topic : Strong entity Set

- A strong entity set is an entity set that contains sufficient attributes to uniquely identify all its entities. In other words, a [redacted] key exists for a strong entity set.
- [redacted] key of a strong entity set is represented by underlining it.



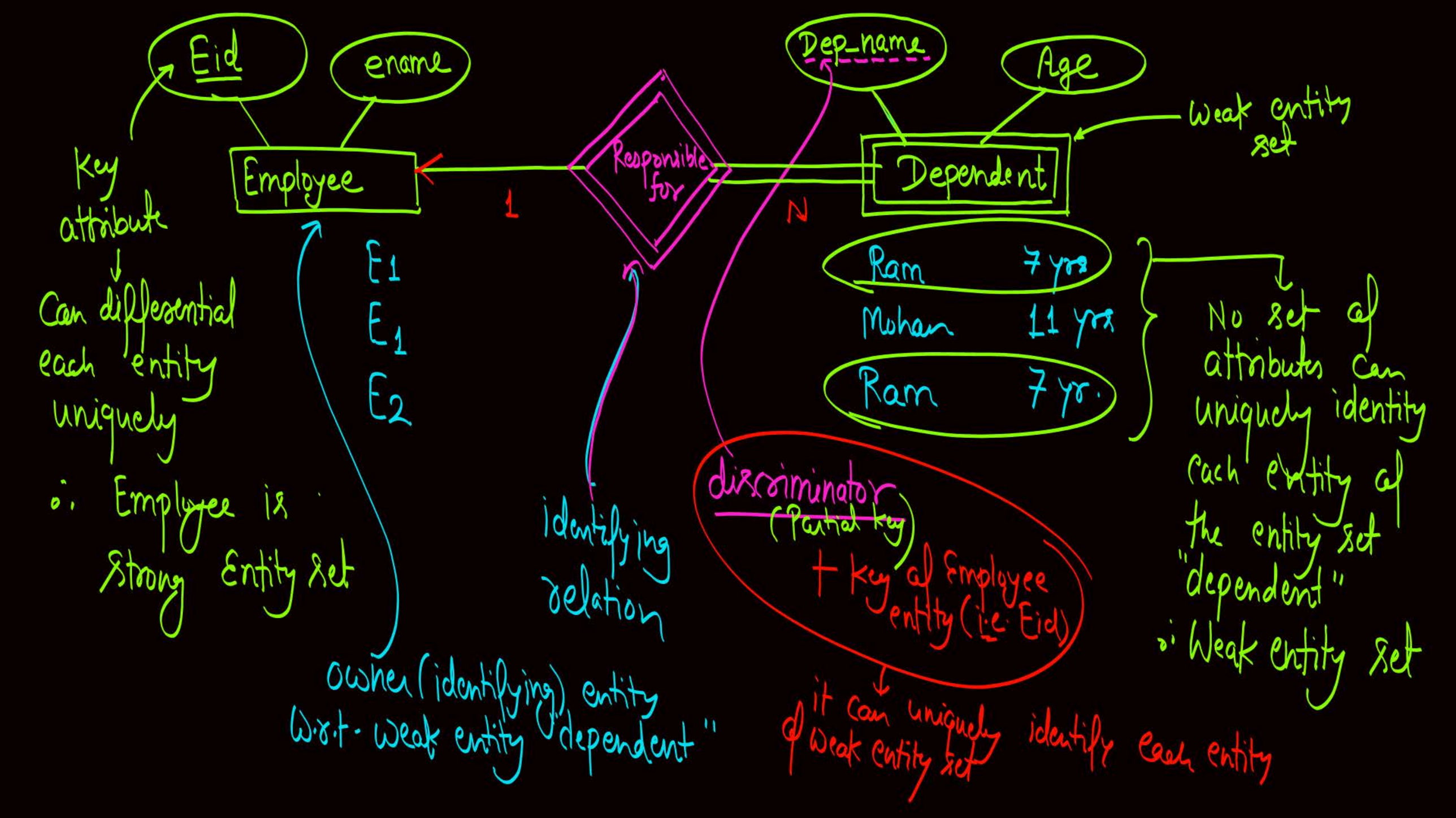
Topic : Weak entity Set

- A weak entity set is an entity set that does not contain sufficient attributes to uniquely identify its entities.
- In other words, a primary key does not exist for a weak entity set.
- However, it contains a partial key called as a discriminator.
- Discriminator can identify a group of entities from the entity set.
- Discriminator is represented by underlining with a dashed line.
- A weak entity must participate in an identifying relationship type with an owner or identifying entity type



Topic : Note

- A weak entity can be identified uniquely only by combining partial key of weak entity set with primary key of owner entity.
- Owner entity set and weak entity set must participate in an one-to-many relationship set. (Owner side is 1 and weak entity side is many)
- Weak entity set must have total participation in identifying relationship set.


Eid
ename
Dep_name
Age
Key attribute
Employee
Responsible for
Dependent
Weak entity set
1
N
Can differential each entity uniquely
∴ Employee is Strong Entity set
E1
E1
E2
Ram 7 yrs
Mohan 11 yrs
Ram 7 yrs.
No set of attributes can uniquely identify each entity of the entity set "dependent"
∴ Weak entity set
identifying relation
discriminator (Partial key)
+ key of Employee entity (i.e. Eid)
owner (identifying) entity W.r.t. weak entity "dependent"
it can uniquely identify each entity of weak entity set

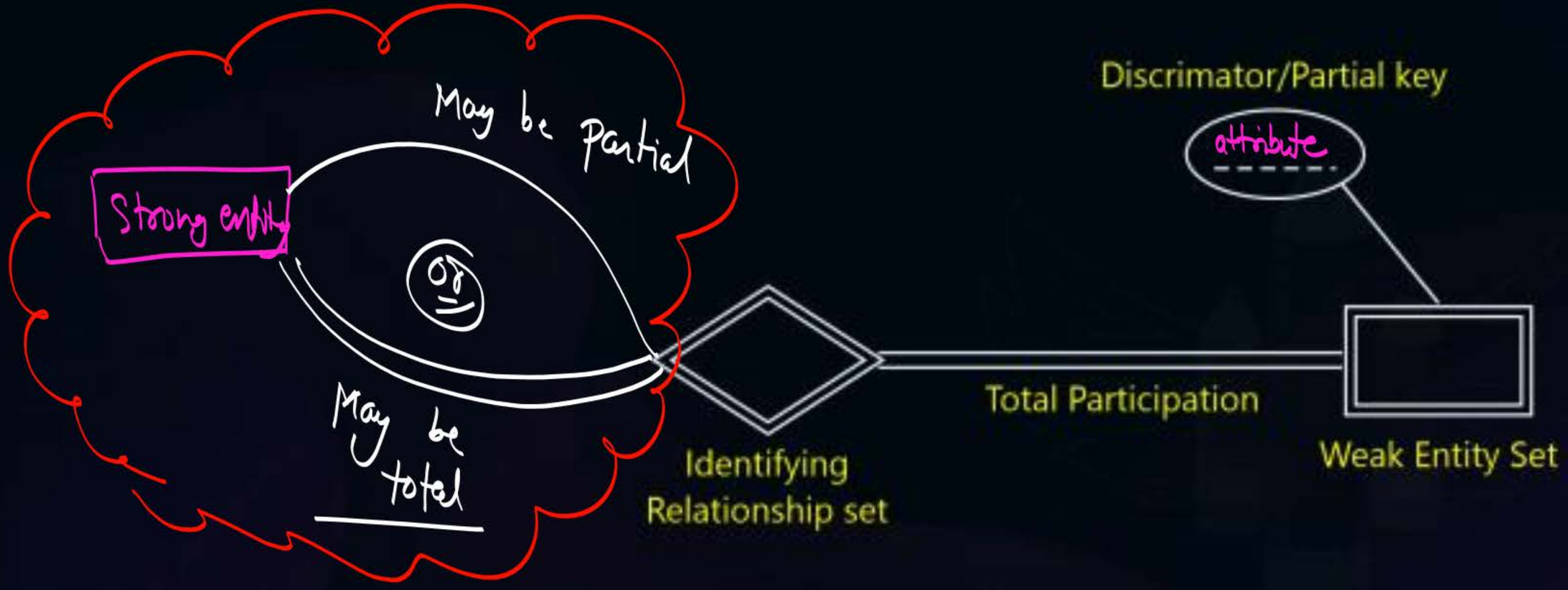


Topic : Differences between Strong entity set and Weak entity set

Strong entity set	Weak entity set
A single line rectangle is used for the representation of a strong entity set.	A double line rectangle is used for the representation of a weak entity set.
It contains sufficient attributes to form its primary key.	It does not contain sufficient attributes to form its primary key.
A single line diamond symbol is used for the representation of the relationship that exists between the two strong entity sets.	A double line diamond symbol is used for the representation of the identifying relationship that exists between the strong and weak entity set.
Total participation may or may not exist in the relationship.	Total participation always exists in the identifying relationship.



Topic : Weak entity set in ER diagram





Topic : Relational Model concept

- The relational model was proposed by E.F. Codd to model data in the form of relations or tables.
- The relational model is represented by a table with columns and rows. Each row is known as a tuple.



Topic : Relational Constraints

These are the restrictions or set of rules imposed on the database content. It validates the quality of the database. It validates that various operations like data insertion, updation, etc., can be performed without affecting the integrity of the data.

Constraints could be either on a column level or a table level. The column level constraints are applied only to one column, whereas the table level constraints are applied to the whole table.



Topic : Types of Constraints on Relational Model

1. Domain constraint
2. Key constraint or Tuple Uniqueness constraint
3. Entity Integrity constraint
4. Referential Integrity constraint

There are many other Constraints, but not important



Topic : Domain constraint

(Column)

Every domain must contain atomic values(smallest indivisible units) which means composite and multi-valued attributes are not allowed.

We perform a datatype check here, which means when we assign a data type to a column we limit the values that it can contain. Eg. If we assign the datatype of attribute age as int, we can't give it values other than int datatype



Topic : Key constraint or Tuple Uniqueness constraint

These are called uniqueness constraints since it ensures that every tuple in the relation should be unique. { i.e. Every relation must have a key }

A relation can have multiple keys or candidate keys(minimal superkey), out of which we choose one of the keys as the primary key, we don't have any restriction on choosing the primary key out of candidate keys.



Topic : Entity Integrity constraint

Entity integrity constraint specifies that no attribute of primary key should contain a null value in any relation.



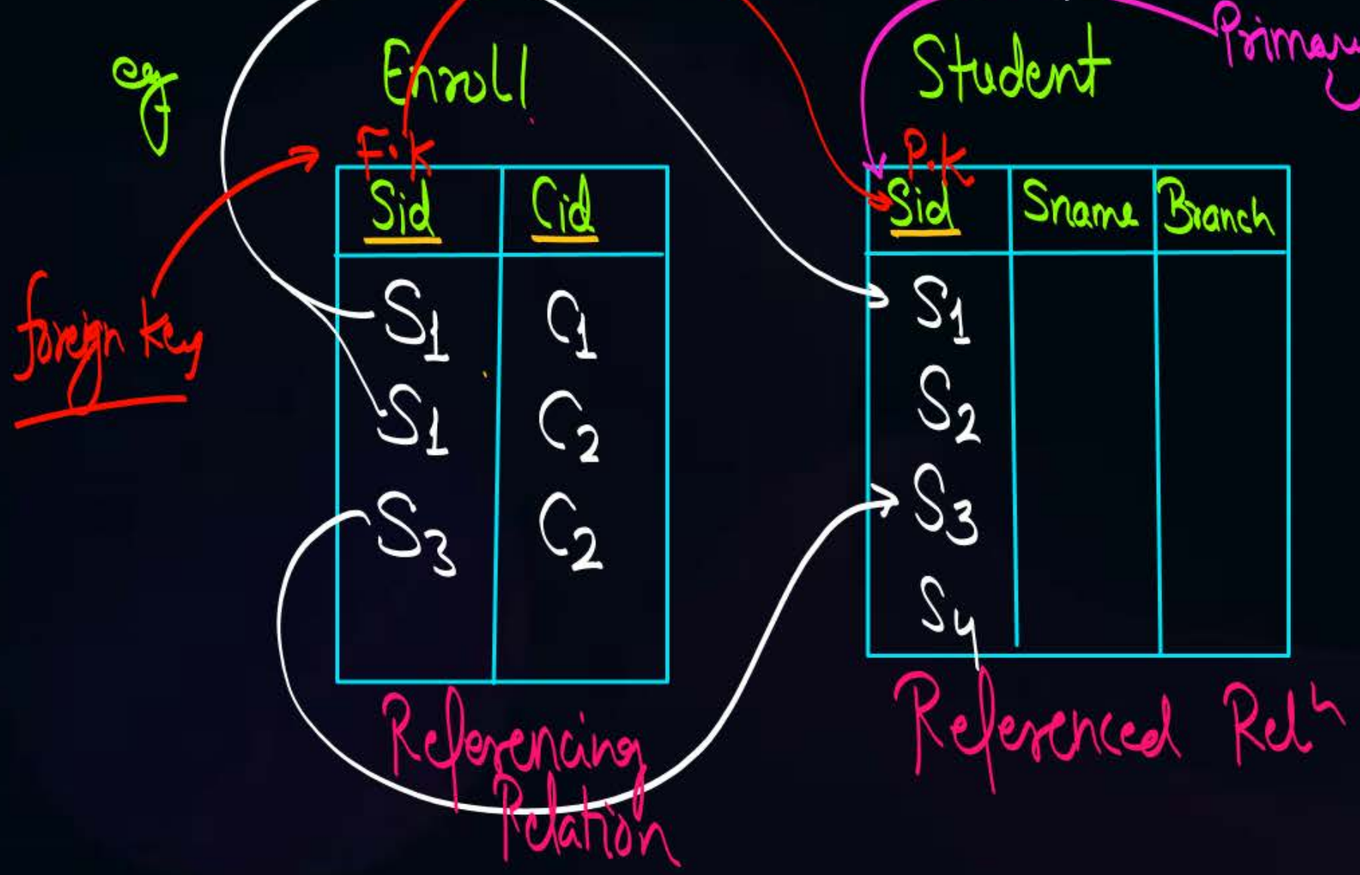
Topic : Referential Integrity constraint

The Referential integrity constraint is specified between two relations or tables and used to maintain the consistency among the tuples in two relations.

Topic : Referential Integrity constraint

it references to the P.K. of Student table

Foreign Key: A foreign key is a set of attributes in a table that refers to the primary key or alternate key of same table or some other table.



{ "Sid" attribute in Enroll table refers to the attribut "Sid" of Student table
{ "Sid" in Student table is Primary Key }



Topic : Referential Integrity constraint

it references to the P.K. of Student table

Foreign Key: A foreign key is a set of attributes in a table that refers to the primary key or alternate key of same table or some other table.

2 Min break

eg

Enroll

F.K Sid	F.K Cid
S_1	C_1
S_1	C_2
S_3	C_2

Foreign keys referring to the Primary or alternate keys of some other table

Referencing Relation

Student

Primary key

P.K Sid	Sname	Branch
S_1		
S_2		
S_3		
S_4		

it is also foreign key referring to Primary key "Cid" of Course relation.

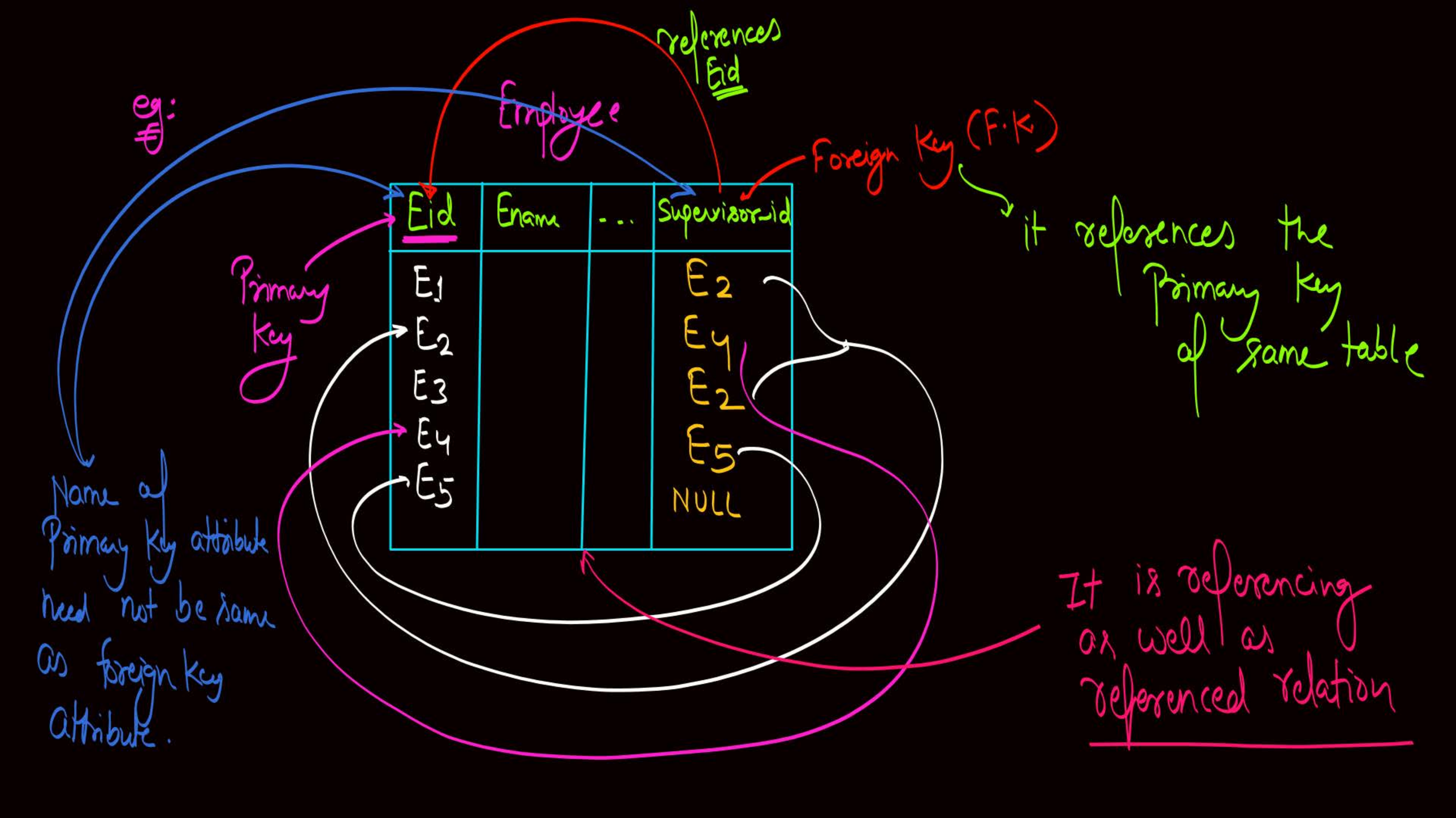
Referenced Relation

Course

Primary key

Cid	Cname
C_1	
C_2	
C_3	

- A relation that contains foreign key is called referencing relation
- Relation that contains primary key (or alternate key) and being referenced by foreign key is called referenced relation


references Eid
eg:
Employee
Foreign key (F.K)
it references the Primary key of same table
Eid
Ename
...
Supervisor_id
Primary key
E1
E2
E3
E4
E5
E2
E4
E2
E5
NULL
Name of Primary key attribute need not be same as foreign key attribute.
It is referencing as well as referenced relation



Topic : Referential Integrity constraint

Referential integrity constraint is enforced through a foreign key

Let foreign key in relation R1 refers to primary key of relation R2.

The values of the foreign key in a tuple of relation R1 can either take the values of the primary key for some tuple in relation R2, or can take NULL values, but can't be empty.



((Values of attribute A') ⊆ values of attribute A)

or NULL



Topic : Referential Integrity constraints on Referenced relation

Insertion: Insertion of a tuple in referenced relation does not result in any referential integrity violation.

{ i.e. Values of foreign key column remain subset of values of Primary key (alternate key) column.

Deletion: Deletion of a tuple from referenced relation may result in referential integrity violation.

In the above eg: if we delete tuple w.r.t. Sid = S_3 from Student rel^n then integrity violation occur in Enroll relation w.r.t. tuple (S_3, C_2).

Updation: Updation of a tuple in referenced relation may result in referential integrity violation.



Topic : Referential Integrity constraints on Referenced relation

(or alternate key)

Deletion of a primary key value from the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

On Delete No Action:

On Delete Cascade:

On Delete Set NULL:



Topic : Referential Integrity constraints on Referenced relation

On Delete No Action: If deletion of a tuple from referenced relation causes any integrity violation, then the deletion of that tuple from referenced relation is prohibited

{i.e No action will be performed}



Topic : Referential Integrity constraints on Referenced relation

On Delete Cascade: If deletion of a tuple from referenced relation causes any integrity violation, then delete the tuple from referenced relation and also delete the tuple from referencing relation w.r.t. integrity violation occur



Topic : Referential Integrity constraints on Referenced relation

On Delete Set NULL: If deletion of a tuple from referenced relation causes any integrity violation and foreign key attribute is allowed to take the 'NULL' value then delete the tuple from referenced relation and set the value of value of foreign key attribute as NULL in the tuples of referencing relation where integrity violation occur.

If deletion of a tuple from referenced relation causes any integrity violation and if foreign key attribute is allowed to take NULL value then deletion from referenced relation itself is prohibited

i.e, If foreign key attribute is not allowed to take NULL value, then "On delete Set NULL" will behave as "ON delete No Action".



Topic : Referential Integrity constraints on Referenced relation

Updation in the primary key (or alternate key) value in the referenced relation may causes integrity violation. We may choose one of the following approaches to avoid integrity violation.

- On Update No Action:
- On Update Cascade:
- On Update Set NULL:



Topic : Referential Integrity constraints on Referenced relation

On Update No Action: If updation in a tuple of referenced relation causes any integrity violation, then the updation in that tuple from referenced relation is prohibited

{ i.e No action will be performed }



Topic : Referential Integrity constraints on Referenced relation

On Update Cascade: If updation of a tuple in referenced relation causes any integrity violation, then update the tuple in referenced relation and also update the tuple in referencing relation w.r.t. integrity violation



Topic : Referential Integrity constraints on Referenced relation

On Update - Set NULL: If updation of a tuple in referenced relation causes any integrity violation and foreign key attribute is allowed to take the 'NULL' value then update the tuple in referenced relation and set the value of value of foreign key attribute as NULL in the tuples of referencing relation where integrity violation occur.

If updation of a tuple in referenced relation causes any integrity violation and if foreign key attribute is allowed to take NULL value then updation in referenced relation itself is prohibited

i.e, If foreign key attribute is not allowed to take NULL value, then "On Update Set NULL" will behave as "ON update No Action".



Topic : Referential Integrity constraints on Referencing relation

Insertion: Insertion of a tuple in referencing relation may result in referential integrity violation.

Deletion: Deletion of a tuple from referencing relation does not result in any referential integrity violation.

Updation: Updation of a tuple in referencing relation may result in referential integrity violation.



Topic : Referential Integrity constraints on Referencing relation

If any operation in referencing relation (child relation) causes any referential integrity violation, then corresponding operation is prohibited.

{we can not disturb parent relation (referenced rel^n)
because of child relation (referencing relation)

① Note:- If R is a relation with 'm' tuples, and S is a relation with 'n' tuples,

then,

No. of tuples in R×S = m.n.

(2) Note: Let R and S are two arbitrary relation

Such that Relation R has 'm' tuples,
and relation S has 'n' tuples.

Then,

Maximum no. of tuples possible in $R \bowtie S = m \cdot n$

Let R(A,B,C) & S(D,E,F)
No Common attribute b/w R&S
$\therefore R \bowtie S = R \times S$

and Minimum number of tuples possible in $R \bowtie S = 0$

let R(AB) & S(BC)

R

A	B
1	2
2	3
3	5

S

B	C
1	2
4	3
6	9

$R \bowtie S = \emptyset$

$\therefore |R \bowtie S| = 0$

③ Note: Let $R(\underline{A}, B, C)$ & $S(\underline{A}, \underline{D}, E)$,

A is Primary Key of R; A & D are Primary Key of S

also No. of tuples in R = m
& No. of tuples in S = n

let attribute 'A' of relation S references to the attribute A of relation R; then find the number of tuples in $R \bowtie S$ = ?

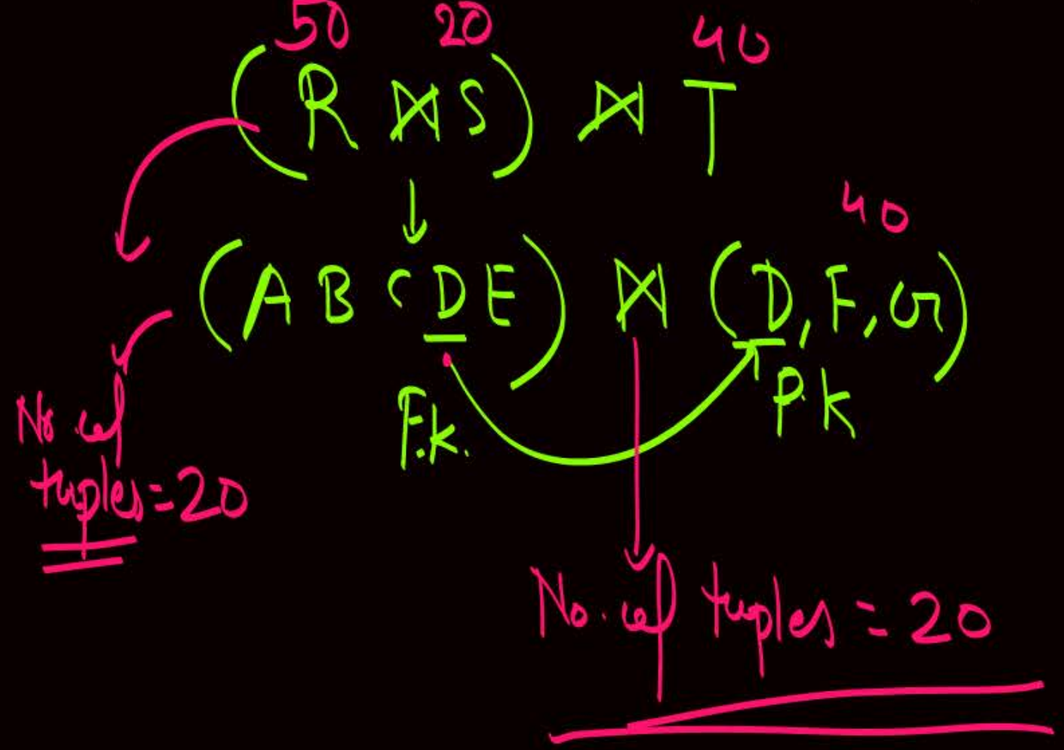
In $R \bowtie S$, each tuple of relation S will match with exactly one tuple of relation R {because value of attribut A in relation R are unique}

∴ In the result of $R \bowtie S$, number of tuples will be exactly same as no. of tuples in relation S, i.e. = "n"

(4) Note:- If R & S are two relations such that R is a referenced relation & S is a referencing rel^n (i.e. relation with foreign key), then in the result of R ⋈ S, number of tuples will always be equal to the number of tuples in relation with foreign key attribute.

Name of the primary key attribute must be same as name of foreign key attribute

Q: Let R(A B C), S(A, D, E), T(D, F, G) are three relations with 50, 20, & 40 tuples respectively. then find the No. of tuples in $(R \bowtie S \bowtie T)$





Topic : ER model to RDBMS table

ER model ⟶ we need ⟶ RDBMS table



2 mins Summary

Topic ER model & ER diagram

Topic Relational model & Integrity constraints

THANK - YOU